

DATE OF ISSUE		28.06.18	09.04.21																	
DRAWING PACKAGE VERSION		1	2																	

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EME EXCLUSION ZONES

LEASE / LICENCE

METAS19-1101201	LEASE DETAIL SURVEY PLAN	0	-																	

DISTRIBUTION

OPTUS	SAFAET HOSSAIN	1	1																	
OPTUS (OPERATIONS)	FOPS	1	1																	
OPTUS VENDOR	IRFAN KHAN (HUAWEI)	1	1																	



OPTUS SITE - H8097

AUSTINS FERRY

TELSTRA CO-LOCATION

PIOMENA RESERVE

2 PLAN 158154 WAKEHURST ROAD

AUSTINS FERRY, TAS 7011

eJV GREENFIELD PROJECT

OPTUS WORK AUTHORITY N° 372842



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AS BUILT

Drawing No.
H8097 - 00

DATE OF ISSUE	28.06.18	09.04.21																		
DRAWING PACKAGE VERSION	1	2																		

REFERENCE DOCUMENTS

OSD-010	OPTUS CONSTRUCTION SPECIFICATION	13	-																
OSD-020	OPTUS EARTHING SPECIFICATION	12	-																
OSD-060	OPTUS ACCESS SPECIFICATION	11	-																
OSD-100	STANDARD CONSTRUCTION NOTES	B	-																
OSD-140	SITE BOUNDARY FENCE & GATE DETAILS	B	-																
OSD-170	SITE SIGNAGE TYPICAL GROUND SITE	B	-																
OSD-191	OPTUS EME SAFETY SIGNAGE REQUIREMENTS	A	-																
OSD-340	PARABOLIC ANTENNA STRAP MOUNTS ON MONOPOLES	A	-																
OSD-510	ELEVATED CABLE LADDER SUPPORT DETAILS SHEET 1	A	-																
OSD-520	ELEVATED CABLE LADDER SUPPORT DETAILS SHEET 2	B	-																
OSD-530	CABLE RISER FIXING TO MONOPOLES	B	-																
OSD-750	TYPICAL GROUND SITE EARTHING DETAILS SHEET 1	B	-																
OSD-760	TYPICAL GROUND SITE EARTHING DETAILS SHEET 2	B	-																
OSD-830	EARTH BAR DETAILS	B	-																
OSD-831	SINGLE POINT EARTH BAR DETAILS	A	-																
STD-HW-005	COLLAR MOUNT DETAILS - SHEET 1 OF 2	A	-																
STD-HW-006	COLLAR MOUNT DETAILS - SHEET 2 OF 2	A	-																
STD-HW-210	OPTUS ZONECOOL SHELTER PIER FOOTING DETAILS	A	-																
MSD-E101	EARTHING POINT CONNECTION - GALVANISED STEEL HOST GRID	02	-																
MSD-E103	EARTHING POINT CONNECTION - CO-LOCATION SITES	02	-																
MSD-S430	FREESTANDING MSB SUPPORTING FRAME	1	-																
H8097-1-7011007	METASITE STRUCTURAL CERTIFICATE DATED 28/06/2018	1	-																
28272-1	ROCLA MONOPOLE CERTIFICATE DATED 01/06/2018	0	-																
5203	GAMCORP FOUNDATION CERTIFICATE DATED 28/05/2018	0	-																
017866P203-11	CO-LOCATE TELSTRA SITE USING MOBILES LOW IMPACT SHELTER	2	-																



OPTUS SITE - H8097

AUSTINS FERRY

TELSTRA SITE ID: 286936

TELSTRA CO-LOCATION

PIOMENA RESERVE

2 PLAN 158154 WAKEHURST ROAD

AUSTINS FERRY, TAS 7011

eJV GREENFIELD PROJECT

OPTUS WORK AUTHORITY N° 372842



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AS BUILT

Drawing No.
H8097 - 00

TELSTRA CO-SITE, PIOMENA RESERVE,
2 PLAN 158154 WAKEHURST ROAD
AUSTINS FERRY, TAS 7011



WGS84 DATUM (USED BY GOOGLE EARTH® AND GPS DEVICES) CAN BE CONSIDERED SAME AS GDA94 (SOURCE: "GEOCENTRIC DATUM OF AUSTRALIA TECHNICAL MANUAL" VERSION 2.3)

1. EXISTING OPTUS 27m HIGH CONCRETE MONOPOLE (SR3-BM27-585). MONOPOLE AND FOUNDATION DESIGN, INCLUDING CERTIFICATION PROVIDED BY OTHERS.
NOTE: POLE PREVIOUSLY OWNED BY TELSTRA. POLE OWNER SHIP HAS NOW BEEN TRANSFERRED TO OPTUS AS PART OF AN OPTUS AND TELSTRA "BUY BACK" AGREEMENT.
2. OPTUS CLAMP ON HEADFRAME TO SUPPORT PANEL ANTENNAS AT EL. 23.9m AND MOUNT TO SUPPORT PARABOLIC ANTENNA AT EL 20.9m.
3. ALL EXISTING OH/UG INFRASTRUCTURE ACCURATELY LOCATED PRIOR TO CONSTRUCTION.
4. ANTENNA MAINTENANCE ACCESS BY QUALIFIED RIGGER PERSONNEL ONLY USING EWP.
5. STRUCTURAL ADEQUACY OF MONOPOLE CONFIRMED BY ROCLA. REFER TO CERTIFICATE 28272-1 DATED 01/06/2018.
6. STRUCTURAL ADEQUACY OF MONOPOLE FOUNDATION CONFIRMED BY ROCLA. REFER TO CERTIFICATE 5203 DATED 28/05/2018.
7. STRUCTURAL ADEQUACY OF HEADFRAME AND PARABOLIC ANTENNA MOUNTS CONFIRMED BY METASITE. REFER TO STRUCTURAL ADEQUACY STATEMENT H8097-1-7011007 DATED 28/06/2018.

1. OPTUS ZONECOOL C3 (3.0m x 2.5m) EQUIPMENT SHELTER.
2. OPTUS SHELTER COLOURED COLORBOND "PALE EUCALYPT".
3. SHELTER SUPPORTED ON STRIP FOOTINGS AND PIERS. REFER TO DRG'S H8097-G4 AND H8097-S1 FOR FOOTING DETAILS.
4. STRUCTURAL ADEQUACY OF PIER FOOTINGS SUPPORT SHELTER CONFIRMED BY METASITE. REFER TO CERTIFICATE H8097-1-1011007 DATED 15/06/2018.

1. THIS SITE LINKED TO THE NETWORK VIA RADIO IN ACCORDANCE WITH DRAWING H8097-T1.
2. WORKS AT THE LINK SITE CARRIED OUT IN ACCORDANCE WITH LINK SITE'S TRANSMISSION DRAWINGS, ISSUED SEPARATELY.

1. FOR IDENTIFIED SITE HAZARD, RISK TREATMENTS AND RESIDUAL RISK FOR THE PROJECT ELEMENTS DESIGNED BY METASITE. REFER TO THE SAFETY IN DESIGN REPORT AND ASSOCIATED RISK REGISTER.
2. PARTIES WHO ACCESS THIS SITE COMPLETE THEIR OWN HAZARD IDENTIFICATION. THE RISK ASSESSMENT AND THE ABOVE REGISTER DOES NOT ALLEVIATE THIS RESPONSIBILITY.

1. ENTER POIMENA RESERVE FROM WAKEHURST ROAD AND TRAVEL TO UPPER CARPARK. OPTUS TO DAISY CHAIN LOCK INTO GATE TO GAIN ACCESS TO GATE.
2. TRAVEL UP EXISTING ACCESS TRACK TO TELSTRA FACILITY.
(NOTE: ACCESS TRACK IS ON/NEXT TO FRISBEE GOLF COURSE, CAUTION SHOULD BE TAKEN)

1. SITE SIGNAGE IS IN ACCORDANCE WITH OSD-170 (GROUND SITE).
2. REFER TO OSD-191 FOR OPTUS EME SAFETY SIGNAGE REQUIREMENTS.

REFER TO EME GUIDE FOR LATEST EME EXCLUSION ZONES FOR ANTENNAS AT THIS SITE.

REFER TO SITE POWER DETAILS DRAWINGS H8097-E1, H8097-E2 AND TO ELECTRICAL ASSET LOCATION ON DRAWINGS H8097-G2 AND H8097-G3.

OPTUS SITE EARTHING COMPLIES WITH OPTUS EARTHING SPECIFICATION DOCUMENT OSD-020, TELSTRA STANDARD DRAWING 016866P203 SHEET 11, METASITE STANDARD DRAWING MSD-E101, MSD-E103 AND THE SITE SPECIFIC EARTHING DRAWING H8097-E3.

AB	09.04.21	AS BUILT	METASITE	AU	JB	GC	PJ
A	28.06.18	ISSUED FOR CONSTRUCTION (EJV GREENFIELD PROJECT)	METASITE	JM	JB	GC	PJ
Rev	Date	Revision Details	Consultant	CAD	Designer	Verifier	Approver



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Subject: MOBILE NETWORK
AUSTRALIA
SITE No:- H8097
AUSTINS FERRY

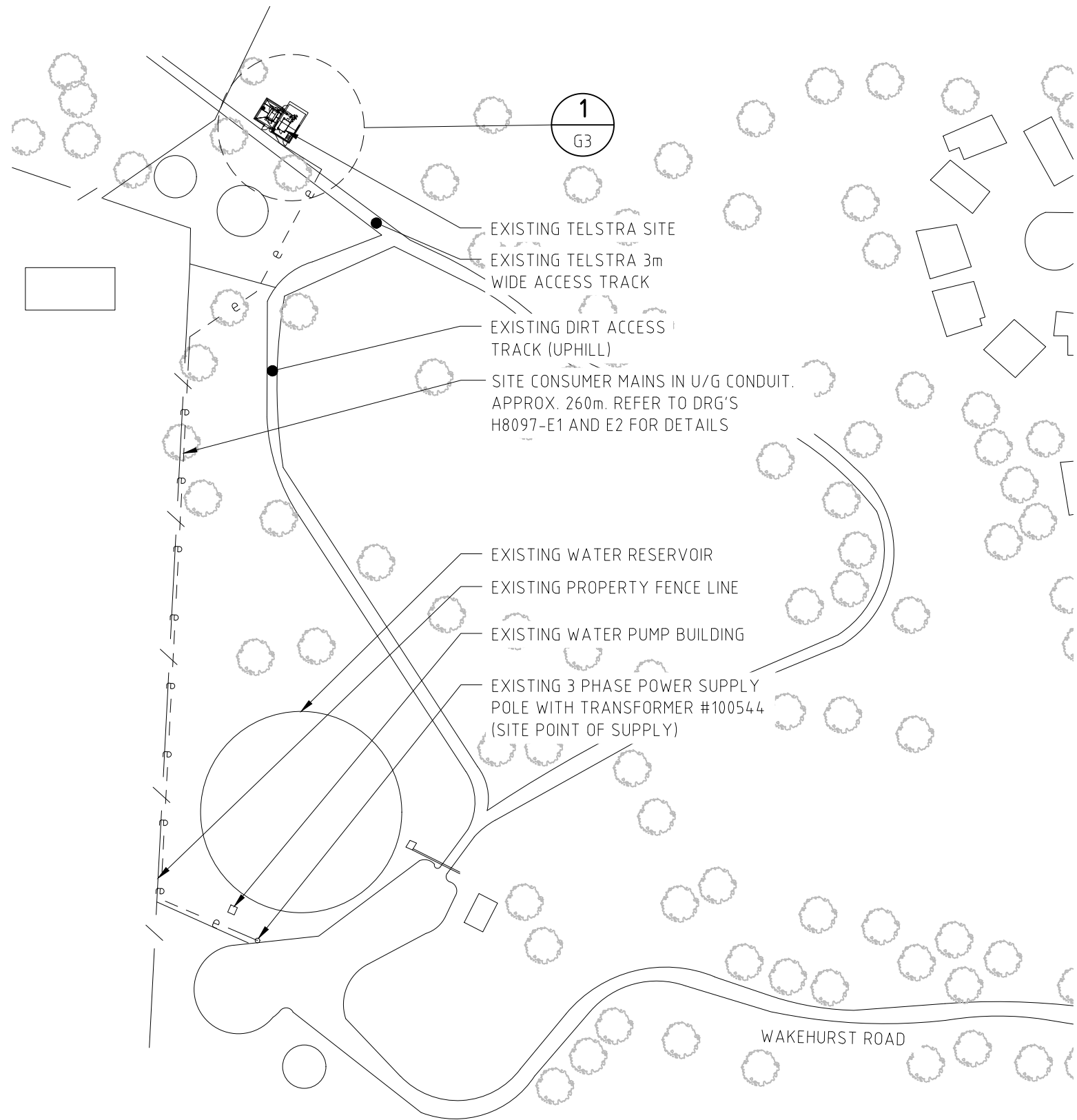
2 PLAN 158154 WAKEHURST ROAD

SITE SPECIFICATIONS

AS BUILT

H8097-G1

Revision
AB



SCALE 1:1500



OPTUS

AB



OPTUS 2.4m HIGH CHAINWIRE SECURITY FENCE WITH 1.5m WIDE SINGLE ACCESS GATE. REFER TO DRAWING OSD-140 FOR DETAILS

OPTUS 75mm THICK SINGLE SIZE GRAVEL/RECYCLED CONCRETE OVER WEED MAT WITH TIMBER EDGING. REFER TO OSD-010 FOR MATERIAL DETAILS AND OSD-140 FOR TIMBER EDGING DETAILS

OPTUS 200mm HIGH TIMBER STEPS. REFER TO DRG. H8097-S1 FOR DETAILS

OPTUS EQUIPMENT SHELTER SUPPORTED ON STRIP FOOTINGS AND PIERS. REFER TO DRG. H8097-S1 FOR FOOTING DETAILS

OPTUS ZONECOOL C3 (3.0m x 2.5m) EQUIPMENT SHELTER. REFER TO DRG'S S2317-G1 AND S2317-F1 FOR SHELTER DETAILS

OPTUS 450mm WIDE 16A NEMA ELEVATED CABLE LADDER WITH COVER AND SUPPORT POSTS. REFER TO DRAWINGS OSD-510 AND OSD-520 FOR DETAILS

EXISTING TELSTRA FENCE CUT LOCALLY FOR OPTUS CABLE LADDER TO PASS THROUGH

OPTUS RRU'S (5-OFF PER SECTOR, 15-OFF TOTAL) AND VHA RRU'S (4-OFF PER SECTOR, 12-OFF TOTAL)

EXISTING OPTUS 27m HIGH CONCRETE MONOPOLE (SR3-BM27-585)

OPTUS COLLAR MOUNT. REFER TO DRAWINGS STD-HW-005 AND STD-HW006 FOR DETAILS

OPTUS Ø600 PARABOLIC ANTENNA (1-OFF) WITH ODU (1-OFF) REMOTE MOUNTED BELOW DISH ON ANTENNA MOUNT. REFER TO DRAWING OSD-340 FOR MOUNT DETAILS

MGA ZONE	55
E	520 051
N	5 263 789
AT	€ MONOPOLE

OPTUS/VHA PANEL ANTENNAS (1-OFF PER SECTOR, 3-OFF TOTAL) WITH OPTUS COMBINERS (3-OFF PER SECTOR, 9-OFF TOTAL) AND VHA COMBINERS (3-OFF PER SECTOR, 9-OFF TOTAL). REFER TO NOTE 7

EXISTING TELSTRA COMPOUND FENCE

EXISTING TELSTRA EQUIPMENT SHELTER

LEGEND

- PROPERTY BOUNDARY
- - - e - - - e - - - U/G CONSUMER MAINS
- 0 e - 0 e - 0 e - OPTUS U/G ELECTRICAL
- / - / - / - FENCE LINE
- + RL XXX.X EXISTING LEVEL
- XXX.X --- EXISTING CONTOUR

ANTENNA LEGEND



NOTES

1. ALL ORIENTATIONS ARE IN DEGREES RELATIVE TO M.G.A. NORTH.
2. REFER TO DRAWING OSD-100 FOR STANDARD CONSTRUCTION NOTES.
3. REFER TO DRAWING H8097-A1 FOR ANTENNA SYSTEM CONFIGURATION.
4. REFER TO DRAWING H8097-T1 FOR SITE TRANSMISSION DETAILS.
5. REFER TO DRAWING H8097-F1 FOR EQUIPMENT SHELTER/OTC LAYOUT.
6. EXISTING OTHER CARRIERS ANTENNAS ARE NOT SHOWN FOR CLARITY.
7. OPTUS AND VHA COMBINERS MOUNTED BEHIND PANEL ANTENNAS AS PER MANUFACTURER'S SPECIFICATIONS.

DETAIL

SCALE 1:100

1
G2



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OPTUS

Client:

Project:

MOBILE NETWORK
AUSTRALIA
SITE No:- H8097
AUSTINS FERRY
2 PLAN 158154 WAKEHURST ROAD

Drawing Title:

SITE LAYOUT AND SETOUT PLAN

Drawing Status:

AS BUILT

Drawing No.

H8097-G3

Revision

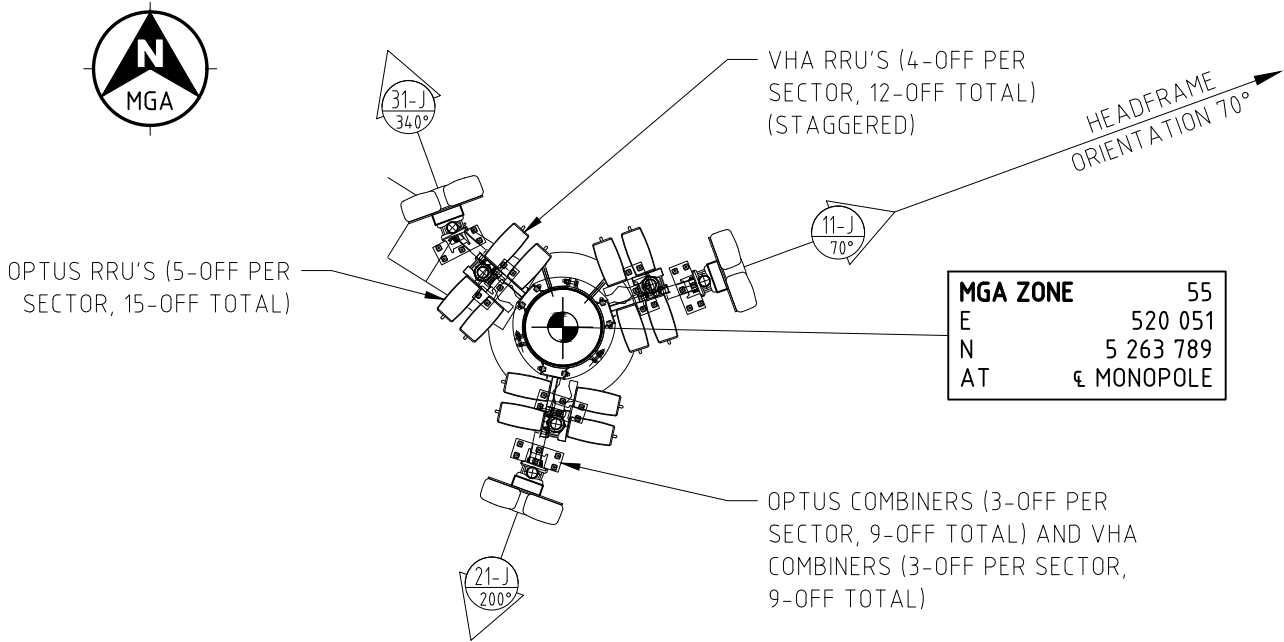
AB

ANTENNA	OPERATOR SECTOR IDENTITY N° (SAO)	OPTUS/VHA SECTOR 1 11-J						OPTUS/VHA SECTOR 2 21-J						OPTUS/VHA SECTOR 3 31-J					
	STATUS	EXISTING						EXISTING						EXISTING					
	AZMUTH (° TN)	70° TN						200° TN						340° TN					
	EL C/L ANTENNA	23.1m						23.1m						23.1m					
	CO-ORDINATES (NOTE 2)	E 520 051			N 5 263 789			E 520 051			N 5 263 789			E 520 051			N 5 263 789		
	MECHANICAL TILT (°)	0°						0°						0°					
	MAKE & MODEL	HUAWEI ASI4517R1						HUAWEI ASI4517R1						HUAWEI ASI4517R1					
	DIMENSIONS (H x W x D)	2600 x 548 x 150						2600 x 548 x 150						2600 x 548 x 150					
	PORTS	1 & 2	3 & 4	5 & 6	7 & 8	9 & 10	11 & 12	1 & 2	3 & 4	5 & 6	7 & 8	9 & 10	11 & 12	1 & 2	3 & 4	5 & 6	7 & 8	9 & 10	11 & 12
	PORT USER	V	O	V	V	O	O	V	O	V	V	O	O	V	O	V	V	O	O
FREQUENCY BANDS	850/900	700/900	1800/2100	1800/2100	18/21/26	18/21/26	850/900	700/900	1800/2100	1800/2100	18/21/26	18/21/26	850/900	700/900	1800/2100	1800/2100	18/21/26	18/21/26	
RET	2°	2°	2°	2°	2°	2°	2°	2°	2°	2°	2°	2°	2°	2°	2°	2°	2°	2°	
ANCILLARIES	MHA	N/A						N/A						N/A					
	COMBINER / DIPLEXER	1 x COM2a (7/9)(O), 2 x COM3 (18/21/23/26)(O), 1 x COM2 (7/8.5/9)(V), 2 x COM4 (18/21)(V)						1 x COM2a (7/9)(O), 2 x COM3 (18/21/23/26)(O), 1 x COM2 (7/8.5/9)(V), 2 x COM4 (18/21)(V)						1 x COM2a (7/9)(O), 2 x COM3 (18/21/23/26)(O), 1 x COM2 (7/8.5/9)(V), 2 x COM4 (18/21)(V)					
	RRU / RF MODULES	1 x RRU3268 (L7)(O), 1 x RRU3936(U9)(O), 1 x RRU3971(L18)(O), 1 x RRU3971(L21)(O), 1 x RRU3281(L26)(O), 1x RRU3952(850)(V), 1x RRU3953(U9)(V), 2x RRU3971(18/21)(V)						1 x RRU3268 (L7)(O), 1 x RRU3936(U9)(O), 1 x RRU3971(L18)(O), 1 x RRU3971(L21)(O), 1 x RRU3281(L26)(O), 1x RRU3952(850)(V), 1x RRU3953(U9)(V), 2x RRU3971(18/21)(V)						1 x RRU3268 (L7)(O), 1 x RRU3936(U9)(O), 1 x RRU3971(L18)(O), 1 x RRU3971(L21)(O), 1 x RRU3281(L26)(O), 1x RRU3952(850)(V), 1x RRU3953(U9)(V), 2x RRU3971(18/21)(V)					
COAXIAL FEEDERS	STATUS	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	TECHNOLOGIES	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	QUANTITY	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	DESIGNATION (SIZE)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	ESTIMATED LENGTH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
TRUNK CABLES	STATUS	EXISTING																	
	TECHNOLOGIES	L7/U9/L18/L21/L26 (OPTUS), 8.5/9/18/21 (VHA)																	
	QUANTITY	3 SHARED ACROSS 3 SECTORS																	
	MAKE & MODEL (SIZE)	H&S / MLEH BP																	
	NO. OF DC / SMF PAIRS	3x 9/18 (1 OPTUS, 1 SHARED, 1 VHA), SHARED ACROSS 3 SECTORS																	
	ESTIMATED LENGTH	40m																	
	OTHER	-																	

ANCILLARY	EQUIPMENT CODE	MANUFACTURER PRODUCT CODE	TECHNOLOGY FREQUENCIES	DIMENSIONS (mm x mm x mm)	WEIGHT (kg)
RRU	HUAWEI RRU	3268	700	400 x 300 x 150	20
		3936	900	400 x 300 x 120	20
		3971	1800	400 x 300 x 120	15
		3971	2100	400 x 300 x 120	15
		3281	2600	400 x 300 x 150	21
	OTHER RRU	3952	850	TBA	TBA
		3953	900	TBA	TBA
		3971	1800/2100	TBA	TBA
COMBINER	COM2	TYT-C003D008 (F2V1)	700/850/900	330 x200 x 130	10
	COM2a	E11F13P25	700/900	165 x253 x 103	6
	COM3	E15V90P56	18/21/23/26	250 x 205 x115	7.6
	COM4	E11F05P82	1800/2100	149 x 145 x117	3.5



- NOTES:
- THIS DRAWING IS READ IN CONJUNCTION WITH RF PLUMBING DIAGRAM DRAWING H8097-A2.
 - ANTENNA CO-ORDINATES ARE SPECIFIED FOR CENTRE OF THE MONOPOLE, TO THE NEAREST METRE.
 - INFORMATION IN THE TABLES SUPPLIED AND VERIFIED BY OPTUS.
 - ANCILLARIES REFER TO ITEMS AT OR NEAR THE ANTENNA.
 - TOTAL TAIL QUANTITY: 120.



OPTUS ANTENNAS PLAN

SCALE 1:50

AB	09.04.21	AS BUILT	METASITE	AU	JB	GC	PJ
A	28.06.18	ISSUED FOR CONSTRUCTION (EJV GREENFIELD PROJECT)	METASITE	JM	JB	GC	PJ
Rev	Date	Revision Details	Consultant	CAD	Designer	Verifier	Approver



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Client:

Project: MOBILE NETWORK
AUSTRALIA
SITE No:- H8097
AUSTINS FERRY
2 PLAN 158154 WAKEHURST ROAD

Drawing Title: PANEL ANTENNA SYSTEM CONFIGURATION		
Drawing Status: AS BUILT	Drawing No. H8097-A1	Revision AB

LEGEND

PANEL PORT

DC STOP

DC PATH

RET PATH

DC SOURCE

AISG BYPASS HT

CROSS CONNECTS

LB FEEDER

OPT HB FEEDER

VHA HB FEEDER

RET CABLE

Tx/Rx Paths

Rx Paths

BAND COLOURS

700 MHz

850 MHz

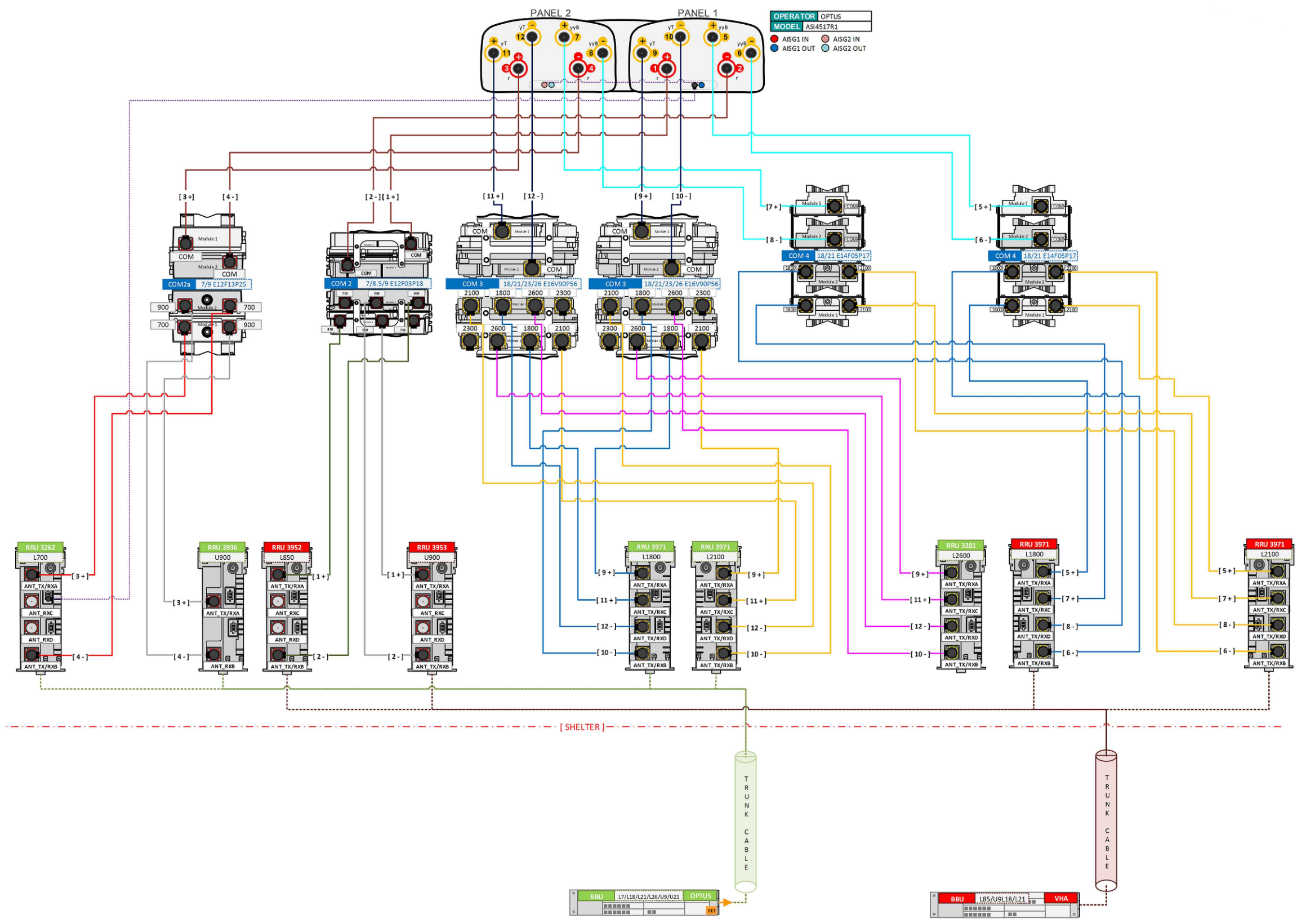
900 MHz

1800 MHz

2100 MHz

2300 MHz

2600 MHz



OPTUS RF PLUMBING DIAGRAM

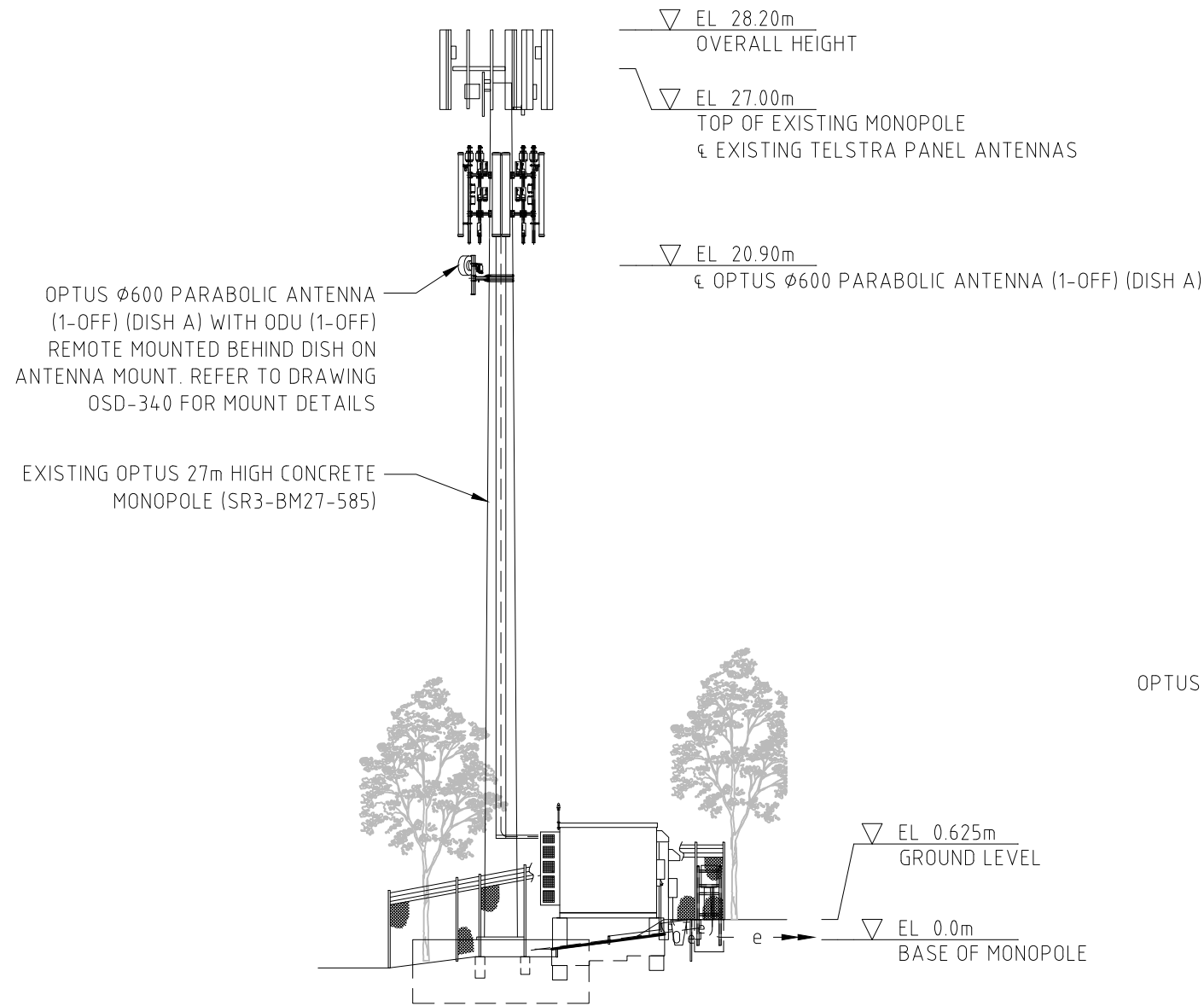
PER SECTOR

- NOTES:**
- THIS DRAWING IS READ IN CONJUNCTION WITH DRG. H8097-A1.
 - RF PLUMBING DIAGRAMS ARE PROVIDED BY OPTUS.

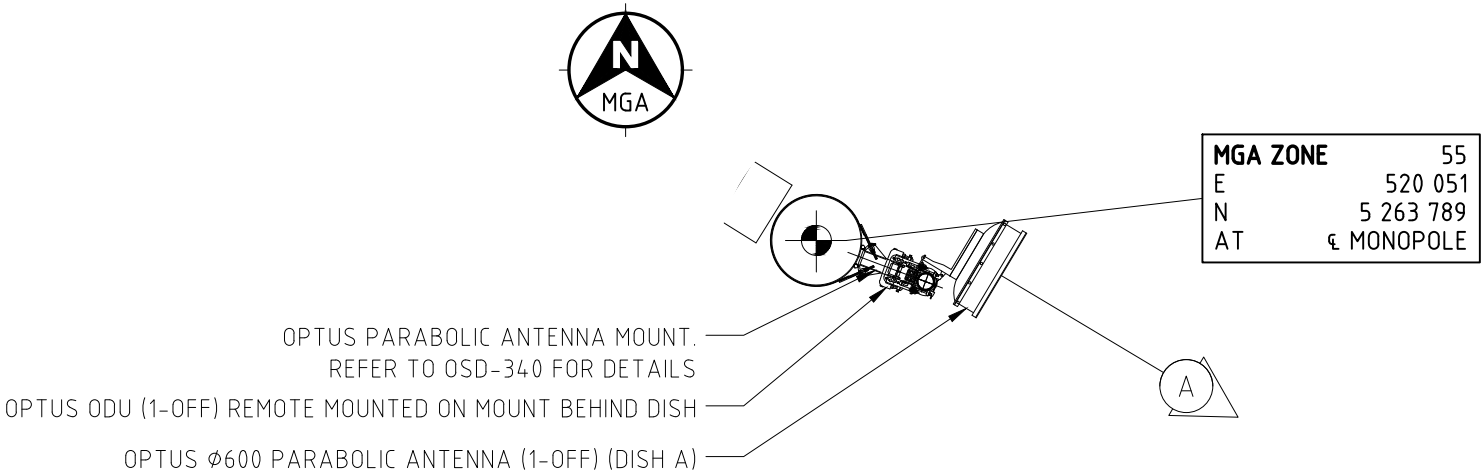
				Client: MOBILE NETWORK AUSTRALIA		Project: SITE No:- H8097 AUSTINS FERRY		Drawing Title: RF PLUMBING DIAGRAM	
HUAWEI TECHNOLOGIES (AU) PTY LTD ABN 49 103 793 380 SYDNEY LEVEL 6 TOWER B 799 PACIFIC HIGHWAY CHATSWOOD NSW 2067 TEL: +61 2 9528 3888 FAX: +61 2 9411 8533 MELBOURNE LEVEL 24 459 COLLINS STREET MELBOURNE VIC 3000 TEL: +61 3 8610 0600 FAX: +61 3 9621 1575				2 PLAN 158154 WAKEHURST ROAD				Drawing Status: AS BUILT	
Revision Details		Consultant CAD Designer Verifier Approver		Drawing No. H8097-A2		Revision AB			

NOTE:
THIS DRAWING IS DIAGRAMMATIC ONLY
AND SHOULD NOT BE SCALED.

TRANSMISSION LINKS - PARABOLIC ANTENNA SCHEDULE							
TAG	STATUS	LINK SITE NAME	SITE No.	AZIMUTH	ELEVATION	DIAMETER	MODEL
A	EXISTING	MT DIRECTION	H0002	120.89° TN	20.9m	ø600	A11D06HS
ODU's							
TAG	STATUS	No.	MODEL	ELEVATION	FEEDER LENGTH	FEEDER DETAILS	
A	EXISTING	1	XMC-2H REMOTE MOUNTED	20.9m	30m	1 x LDF4-50 (1/2")	

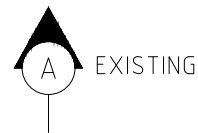


NORTH WEST ELEVATION
SCALE 1:200



OPTUS PARABOLIC ANTENNA PLAN
SCALE 1:50

LEGEND:



NOTES:

1. ALL ANTENNA AZIMUTHS ARE IN DEGREES RELATIVE TO TRUE NORTH.
2. EXISTING OTHER CARRIER ANTENNAS ARE OMITTED FROM THE ANTENNA PLAN FOR CLARITY.
3. THIS DRAWING IS DIAGRAMMATIC ONLY AND SHOULD NOT BE SCALED.
4. OPTUS ø600 PARABOLIC ANTENNA (1-OFF) AND ODU (1-OFF) REMOTE MOUNTED.

AB	09.04.21	AS BUILT	METASITE	AU	JB	GC	PJ
A	28.06.18	ISSUED FOR CONSTRUCTION (EJV GREENFIELD PROJECT)	METASITE	JM	JB	GC	PJ
Rev	Date	Revision Details	Consultant	CAD	Designer	Verifier	Approver



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Client:

OPTUS

Project:

**MOBILE NETWORK
AUSTRALIA
SITE No:- H8097
AUSTINS FERRY
2 PLAN 158154 WAKEHURST ROAD**

Drawing Title:

SITE TRANSMISSION DETAILS

Drawing Status:

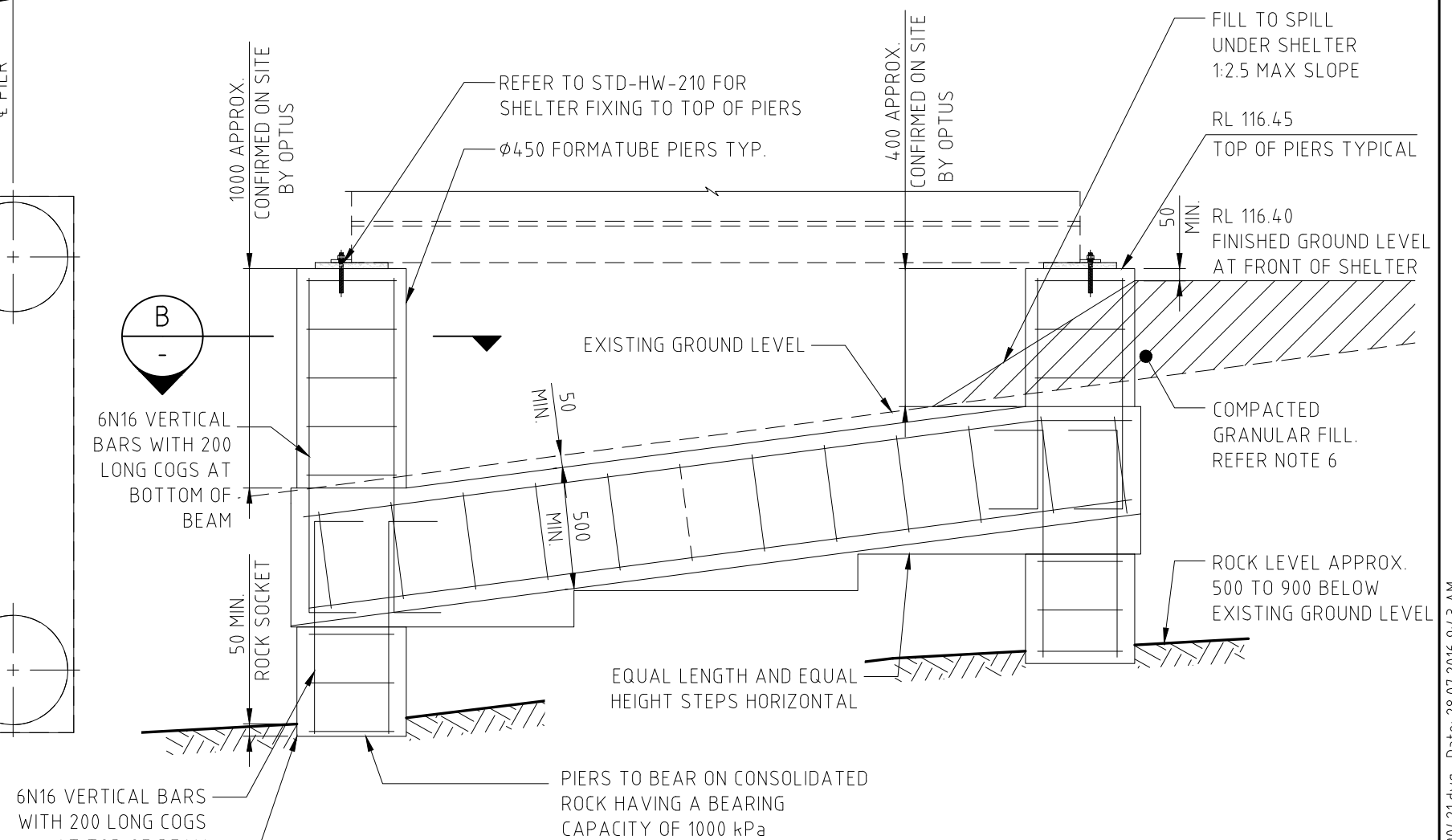
AS BUILT

Drawing No.

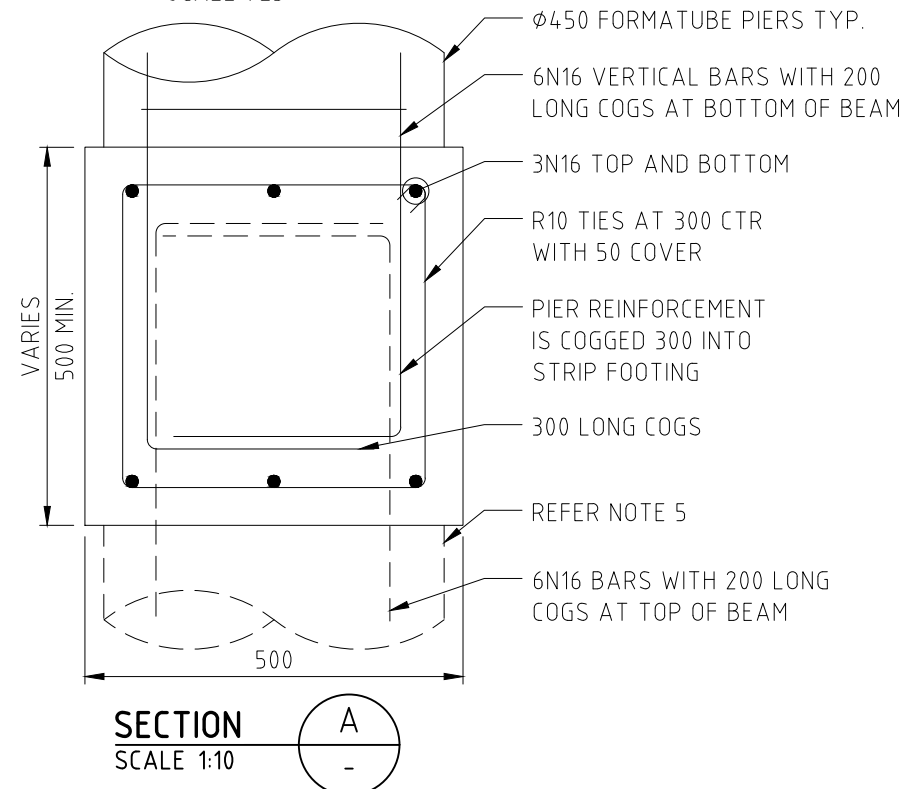
H8097-T1

Revision

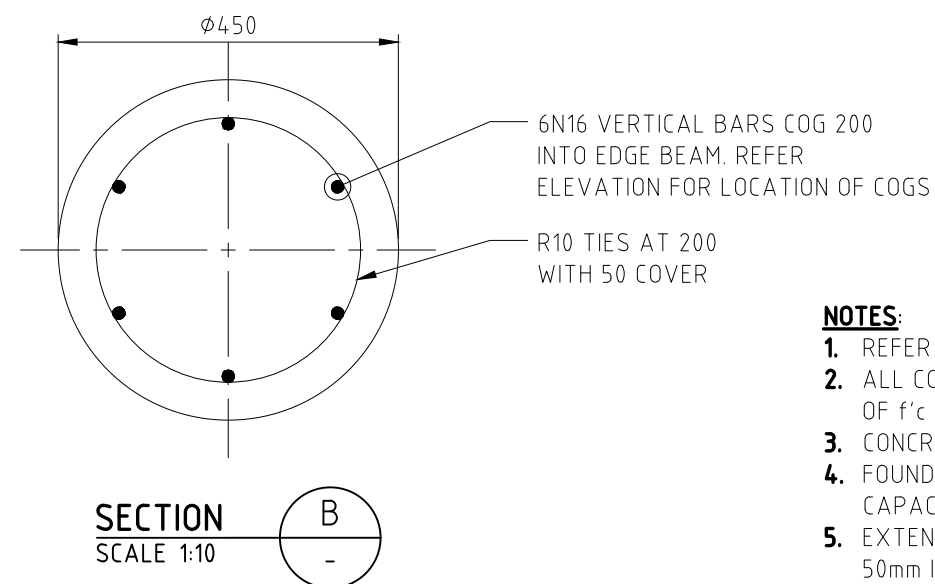
AB



SCALE 1:25



FOUNDATION ELEVATION FB1, FB2, FB3
SCALE 1:25 AND FB4 SIMILAR



NOTES:

1. REFER TO DRG. H8097-G3 FOR SHELTER SET-OUT DETAILS.
2. ALL CONCRETE WORK IN ACCORDANCE WITH AS 3600 WITH CONCRETE STRENGTH OF $f'c$ 32MPa AT 28 DAYS.
3. CONCRETE COVER TO REINFORCEMENT: 50mm TOP; 50mm BOTTOM AND SIDES.
4. FOUNDATION MATERIAL FOR STRIP FOOTINGS TO HAVE A MINIMUM BEARING CAPACITY OF 100 kPa.
5. EXTEND ALL PIERS AND REINFORCEMENT DOWN TO ROCK. PIERS TO SOCKETED 50mm INTO ROCK.
6. FILL GRANULAR MATERIAL COMPACTED TO 95% STANDARD DRY DENSITY.

AB	09.04.21	AS BUILT		METASITE	AU	JB	GC	PJ	
A	28.06.18	ISSUED FOR CONSTRUCTION (EJV GREENFIELD PROJECT)		METASITE	JM	JB	GC	PJ	
Rev	Date	Revision Details		Consultant	CAD Designer		Verifier	Approver	



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Client:	
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OPTUS

Project:

MOBILE NETWORK
AUSTRALIA
SITE No:- H8097
AUSTINS FERRY

2 PLAN 158154 WAKEHURST ROAD

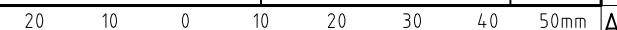
Drawing Title:

SHELTER FOUNDATION DETAILS

Drawing Status:
AS BUILT

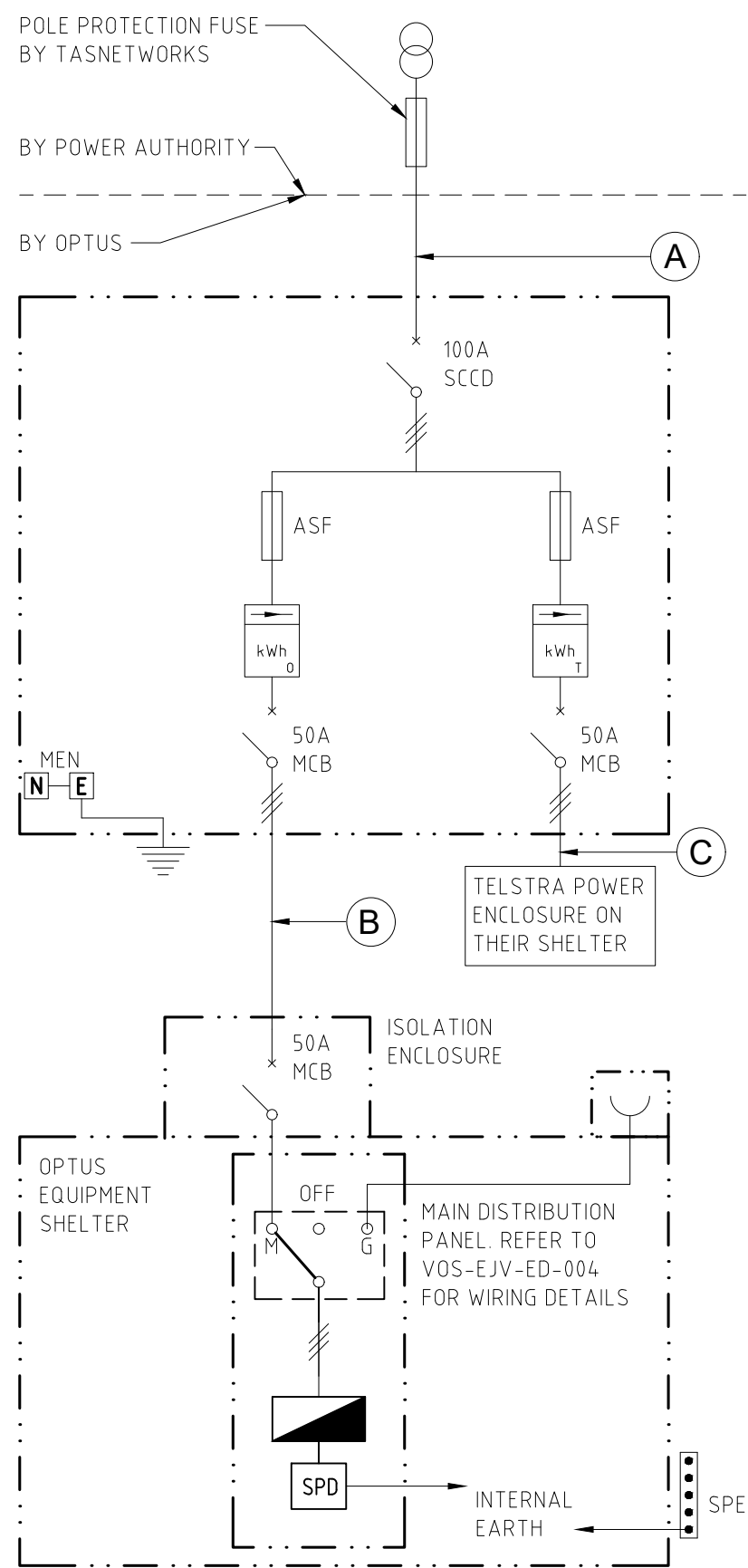
Drawing No.
H8097-S1

Revision
AB





Drawing Title:		
TIMBER STEP DETAIL		
Drawing Status:	Drawing No.	Revision
AS BUILT	H8097-S2	AB



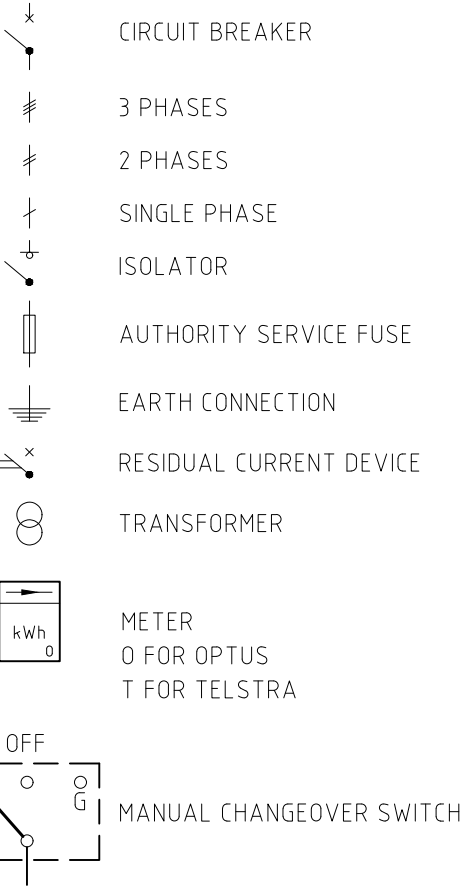
SINGLE LINE DIAGRAM

NTS

CABLE SCHEDULE								
CABLE	CABLE TYPE	CORES	ACTIVE	NEUTRAL	EARTH	CABLE LENGTH	INSTALLATION METHOD	COMMENTS
(A)	XLPE/PVC Cu	4 x 1C	70mm ²	70mm ²	-	260m	U/G IN Ø63mm CONDUIT	BY OPTUS
(B)	XLPE/Cu/SDI	1 x 4C + E	16mm ²	16mm ²	6mm ²	10m	U/G IN P63 PVC CONDUIT	BY OPTUS
(C)	XLPE/Cu	1 x 4C + E	16mm ²	16mm ²	6mm ²	5m	U/G IN Ø63mm CONDUIT	BY OPTUS

LEGEND

N NEUTRAL
E EARTH
MEN MULTIPLE EARTHED NEUTRAL
SPD SURGE PROTECTION DEVICE
SCCD SITE CURRENT CONTROL DEVICE



- NOTES:
- THIS DRAWING READ IN CONJUNCTION WITH H8097-E1.
 - CABLE LENGTHS GIVEN ARE INDICATIVE ONLY. EXACT CABLE ROUTES AND LENGTHS CONFIRMED ON SITE PRIOR TO CONSTRUCTION.



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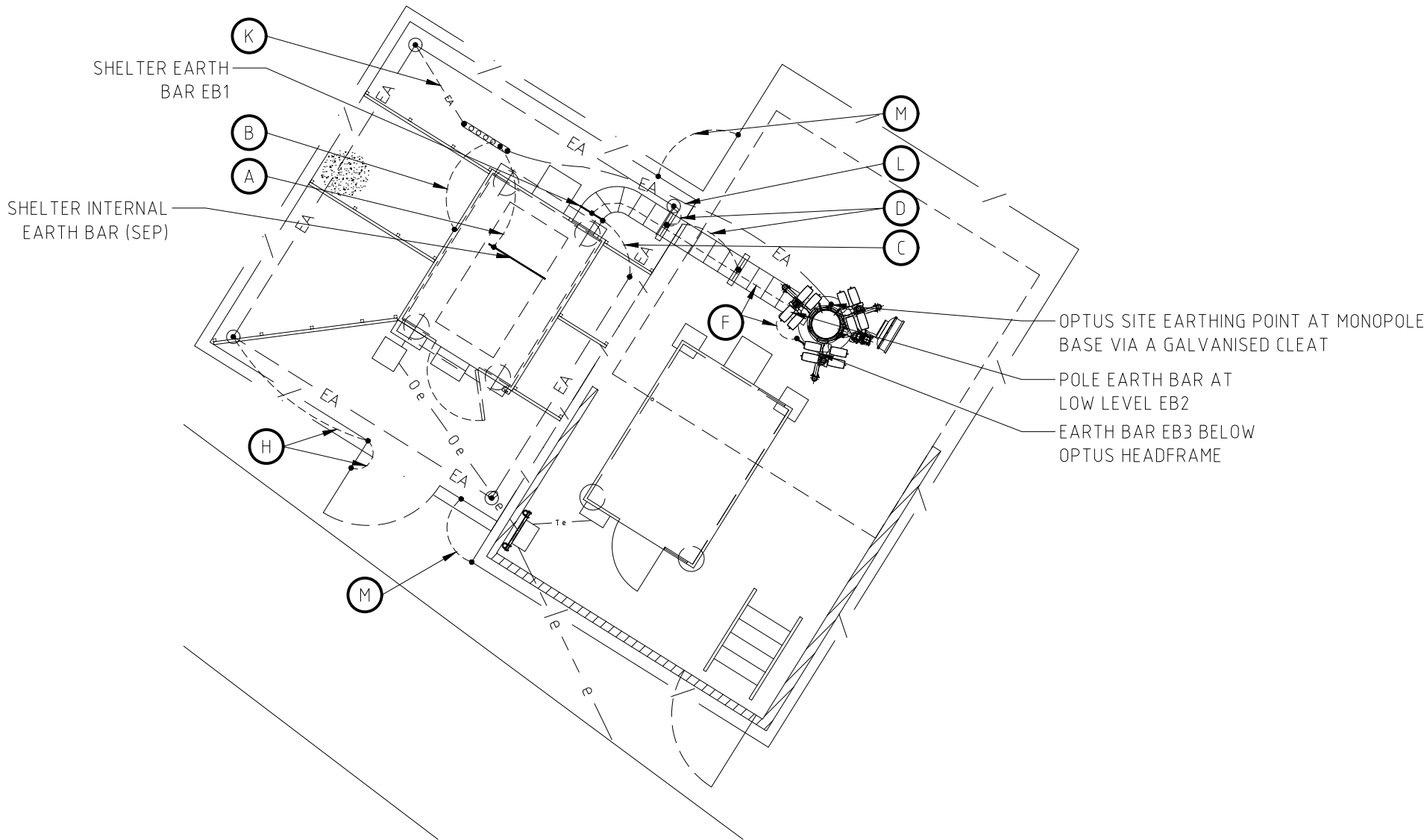
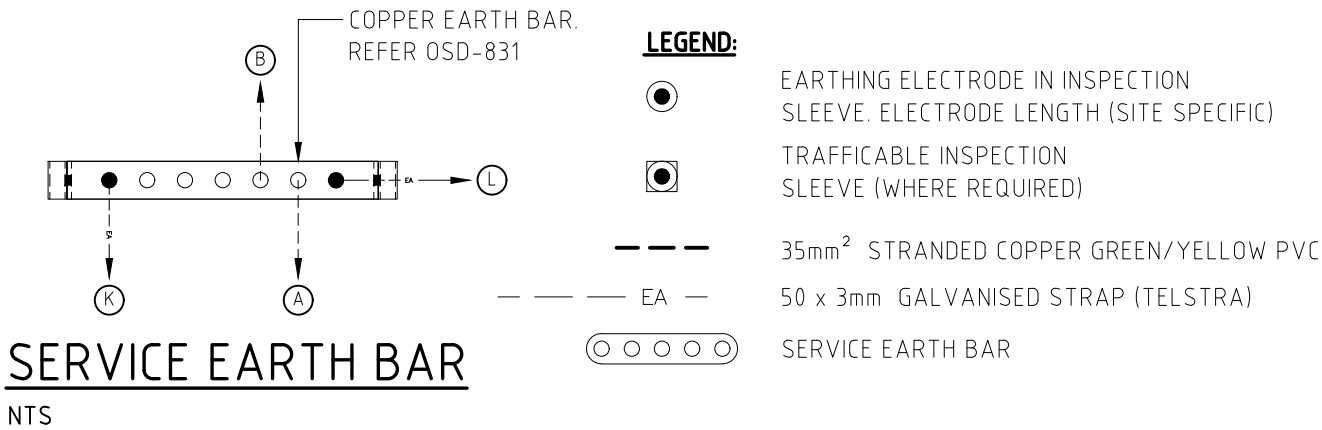
Client:

Project: MOBILE NETWORK AUSTRALIA
SITE No:- H8097
AUSTINS FERRY
2 PLAN 158154 WAKEHURST ROAD

Drawing Title:
SINGLE LINE DIAGRAM

Drawing Status:
AS BUILT

Drawing No. H8097-E2
Revision AB



EARTHING INSTALLATION

ALL ITEMS EARTHED GENERALLY AS SHOWN ON THIS DRAWING. FOR CONNECTION REFERENCE NUMBER, CABLE SIZE AND CONNECTION DETAILS, REFER TO TABLE 1 OF THE OPTUS EARTHING SPECIFICATION (OSD-020).

ITEM	DESCRIPTION	CONNECTION REF No.
A	INTERIOR EARTH TAPE TO SPE	EC4
B	SHELTER BASE FRAME TO SPE	EC2
C	EARTH BAR	EC5
D	CABLE LADDER SUPPORT POST	EC6
F	CABLE TRAY EQUIPOTENTIAL EARTH CABLES BETWEEN EARTH BARS	EC2
H	FENCING AND GATES	EC6/EC9
K	SPE TO ELECTRODE	EC7
L	SPE TO TOWER LEG	-
M	EARTH CONTINUING BETWEEN FENCES FRAMES	EC2

- NOTES:**
- OPTUS EARTH SYSTEM IS OF THE SAME MATERIAL AS HOST STRUCTURE EARTHING SYSTEM. 3 x 50mm HOT DIPPED GALVANISED EARTH STRAPS.
- A MINIMUM OF 4-OFF 13mm DIAMETER SOLID STAINLESS STEEL EARTHING ELECTRODES DRIVEN 3m DEPTH UNLESS SPECIFIED OTHERWISE.
 - ELECTRODES IN 75mm MINIMUM DIAMETER HOLES AND BACKFILL AS PER OSD-020.
 - A MINIMUM OF 3-OFF EARTH BARS AT TOP OF MONOPOLE, ON MONOPOLE ABOVE ELEVATED CABLE LADDER AND THE SHELTER.
 - SINGLE POINT EARTH BAR FACE FIXED TO SHELTER BASE FRAME IN ACCORDANCE WITH OSD-831.
 - SHELTER INTERIOR EARTHING TAPE IS SHELTER SUPPLIER.
 - MECHANICAL EXTERNAL CONNECTIONS ARE TREATED WITH DENSO PASTE AND TAPE.
 - ANY ALTERED SURFACE IN PARENT GRID COMPOUND AS A RESULT OF THIS INSTALLATION REINSTATED TO ITS ORIGINAL CONDITION.
 - MEN LOCATED AT THE METERING POINT.
 - MONOPOLE BOLT ALLOWING FOR OPTUS EARTHING SYSTEM ISOLATION FOR TESTING PURPOSES. REFER TO STD-HW-E001.
 - OPTUS CONNECTION CLEARLY LABELLED AS "OPTUS SITE EARTHING POINT".
 - OPTUS FEEDER EARTH BAR AT THE BASE OF MONOPOLE MOUNTED ON THE CABLE LADDER AS CLOSE POSSIBLE TO FEEDER ENTRY.

Rev	Date	Revision Details	Consultant	CAD	Designer	Verifier	Approver
AB	09.04.21	AS BUILT	METASITE	AU	JB	GC	PJ
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Client:



Project:

MOBILE NETWORK
AUSTRALIA
SITE No:- H8097
AUSTINS FERRY
2 PLAN 158154 WAKEHURST ROAD

Drawing Title:

SITE EARTHING PLAN

Drawing Status:

AS BUILT

Drawing No.

H8097-E3

Revision

AB

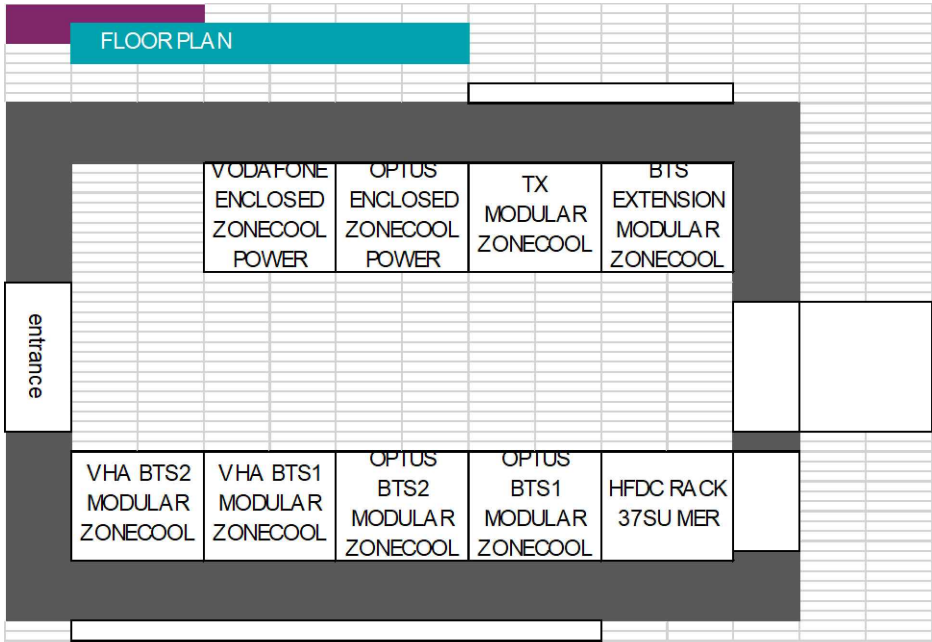
SHELTER DETAILS:

- 1. INTERNATIONAL COMMUNICATIONS SHELTERS (ICS) ZONECOOL TYPE C3 SANDWICH PANEL SHELTER, COLOURED COLORBOND "PALE EUCALYPT".
- 2. SHELTER SELECTION IS BASED ON OSD-010 V13.0 SECTION 3.1.2.
- 3. SHELTER LAYOUT BASED ON ICS DRAWING PACK ISSUE DATE 14-10-2016.
- 4. REFER TO REFER TO DRG H8097-G3 FOR METER BOX LOCATION.
- 5. THIS DRAWING IS DIAGRAMMATIC ONLY AND SHOULD NOT BE SCALED.

DC POWER SUPPLY:

OPTUS AND VHA HUAWEI DC POWER SYSTEM, EACH EQUIPPED WITH 6 x OFF 3kW RECTIFIERS AND 5 x 100AH BATTERY STRINGS.

VODAFONE DC POWER RACK				OPTUS DC POWER RACK				TX RACK				MISC/SPARE RACK			
Modular Zone Cool Rack-Type 1				Modular Zone Cool Rack-Type 1				Modular Zone Cool Rack-Type 2				Modular Zone Cool Rack-Type 2			
44	POWER SYSTEM UNIT			44	POWER SYSTEM UNIT			44				44			
43				43				43	DC FAN INLET CHANNEL			43	DC FAN INLET CHANNEL		
42				42				42				42			
41	MODEL: HUAWEI			41	MODEL: HUAWEI			41				41			
40				40				40				40			
39	6 x 3KW RECTIFIERS			39	6 x 3KW RECTIFIERS			39	RESERVED FOR AIR GUIDE SHELF			39	RESERVED FOR AIR GUIDE SHELF		
38				38				38	Optus Eltek APDP			38	SPARE		
37				37				37	SPARE			37	SPARE		
36				36				36	SPARE			36	SPARE		
35				35				35	SPARE			35	SPARE		
34				34				34	SPARE			34	SPARE		
33	Battery String 5			33	Battery String 5			33	SPARE			33	SPARE		
32				32				32	SPARE			32	SPARE		
31				31				31	SPARE			31	SPARE		
30				30				30	Fibre Management - OPTUS			30	SPARE		
29				29				29	SPARE			29	SPARE		
28				28				28	Optus - Huawei ATN 910			28	SPARE		
27				27				27	SPARE			27	SPARE		
26	Battery String 4			26	Battery String 4			26	Fibre Management - OPTUS			26	SPARE		
25				25				25	SPARE			25	SPARE		
24				24				24	Optus - Huawei RTN 905			24	SPARE		
23				23				23	SPARE			23	SPARE		
22				22				22	SPARE			22	SPARE		
21				21				21	SPARE			21	SPARE		
20				20				20	SPARE			20	SPARE		
19				19				19	SPARE			19	SPARE		
18	Battery String 3			18	Battery String 3			18	SPARE			18	SPARE		
17				17				17	SPARE			17	SPARE		
16				16				16	SPARE			16	SPARE		
15				15				15	SPARE			15	SPARE		
14				14				14	KRONE 1 - 15 Way			14	SPARE		
13				13				13	ENV ALARMS BTS, 1642 or MSR			13	SPARE		
12				12				12	SPARE			12	SPARE		
11	Battery String 2			11	Battery String 2			11	SPARE			11	SPARE		
10				10				10	SPARE			10	SPARE		
9				9				9	SPARE			9	SPARE		
8				8				8	SPARE			8	SPARE		
7				7				7	SPARE			7	SPARE		
6				6				6	SPARE			6	SPARE		
5				5				5				5			
4	Battery String 1			4	Battery String 1			4				4			
3				3				3				3			
2				2				2	AIR INLET VENT			2	AIR INLET VENT		
1				1				1				1			



OPTUS SHELTER
RACK CONFIGURATION (MZC-001)
NTS

AB	09.04.21	AS BUILT	METASITE	AU	JB	GC	PJ
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Client:



Project:

MOBILE NETWORK
AUSTRALIA
SITE No:- H8097
AUSTINS FERRY
2 PLAN 158154 WAKEHURST ROAD

Drawing Title:

EQUIPMENT SHELTER
LAYOUT PLAN - SHEET 1 OF 2

Drawing Status:

AS BUILT

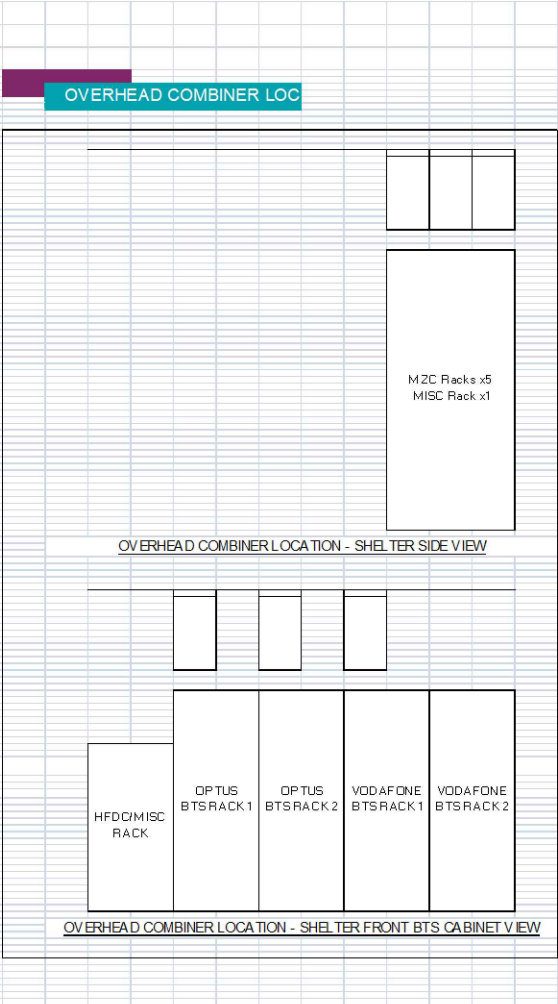
Drawing No.

H8097-F1

Revision

AB

HFDC RACK 37 SU Rack				OPTUS BTS RACK 1 Modular Zone Cool Rack-Type 2				OPTUS BTS RACK 2 Modular Zone Cool Rack-Type 2				VODAFONE BTS RACK 1 Modular Zone Cool Rack-Type 2				VODAFONE BTS RACK 2 Modular Zone Cool Rack-Type 2			
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				42		42	42			42	42			42	42			42	
				41		41	41			41	41			41	41			41	
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34				34	34		SPARE	34	34		SPARE	34	34		SPARE	34	34		SPARE
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32				32	32		SPARE	32	32		SPARE	32	32		SPARE	32	32		SPARE
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30			Optus Hybrid Fibre Termination (H+S)	30	30		SPARE	30	30		SPARE	30	30		SPARE	30	30		SPARE
29			Optus Hybrid Fibre Management Tray (H+S)	29	29		SPARE	29	29		SPARE	29	29		SPARE	29	29		SPARE
28			Optus Hybrid Fibre Termination (H+S)	28	28		SPARE	28	28		SPARE	28	28		SPARE	28	28		SPARE
27			Optus Hybrid Fibre Management Tray (H+S)	27	27		SPARE	27	27		SPARE	27	27		SPARE	27	27		SPARE
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OPTUS SHELTER
RACK CONFIGURATION (MZC-001)
NTS

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MOBILE NETWORK
AUSTRALIA
SITE No:- H8097
AUSTINS FERRY
2 PLAN 158154 WAKEHURST ROAD

Drawing Title:

EQUIPMENT SHELTER
LAYOUT PLAN - SHEET 2 OF 2

Drawing Status:

AS BUILT

Drawing No.

H8097-F1-1

Revision

AB